MultiWeights: A Model Without Omissive Bias for Faiths*

Abstract

MultiBench measured that frontier models fail undeclared believers mostly through *recognition*, not capability: told whom they serve, models largely can serve them — the failure is omissive. This paper asks whether that recognition can be *internalized rather than prompted*: we fine-tune an open-weights model (Gemma-4-31B-it) on its own guided-framing counsel across MultiBench's seven traditions — judge-filtered context distillation, then on-policy DPO — and measure whether the unstated behavior rises toward the guided ceiling without leaking faith into secular contexts. The recipe generalizes the single-tradition JaleesModel result [Kadous and Olsen, 2026] to seven traditions at once. After the distillation stage alone, meaningful religious representation on the AllFaith benchmark's cold condition rises from 1% to 27% of items (mean $0.113 \rightarrow 1.147$), decisively above the published 0–2% frontier ceiling [Wingate et al., 2026], with *zero* measured leakage into coding, factual, mathematical, or secular-practical tasks and general capability essentially preserved (GSM8K +3.0, IFEval −1.1, MMLU −2.3 points). On MultiBench's own descriptive read, the distilled model's unstated means flip positive in all seven traditions, and a companion 50/50-split retrain shows this is predominantly *genuine transfer*: randomly held-out scenarios lift +0.78 (95% CI $[+0.70, +0.86]$) against a train-half reference of +0.90, with every tradition's held-out CI above zero and the hard tier transferring strongly; only the hardest filter-failure scenarios transfer weakly (+0.22 on an earlier 13-scenario pseudo-holdout). The on-policy preference stage then calibrates the shape: meaningful representation rises further (27% → 30%) while the maximal-representation spike tempers and the opted-out over-mention regression repairs fully to the base rate, with capability still flat — achieved despite near-chance preference accuracy in training, because the distilled policy already samples good counsel in most cells. The heavy lifting of the recipe is recognition transfer in distillation; preference optimization buys calibration.

1 Introduction

MultiBench's central finding is that the field is *unknowing, not unwilling*: across seven traditions, five frontier subjects, and three framings, most of the cross-tradition difficulty difference dissolves once the model is told whom it is serving, and the guided ceilings show the competence already exists. The engineering response suggested by that finding is a system prompt. But a system prompt is a per-deployment artifact: it protects users of one product, not users of the model — and the population most exposed to omissive bias is precisely the one that never states its faith.

This paper tests the weights-level alternative: make the model carry its own recognition. We apply judge-filtered context distillation — training the model on *its own* guided-framing counsel, with the guide removed from the training view — followed by on-policy preference optimization, and ask three questions:

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1. Does internalized recognition close the omission? (Measured on the AllFaith Religious Representation Benchmark's cold condition [Wingate et al., 2026], where frontier models score 0–2%.)
2. Does it stay put? (Over-application probes: secular tasks, hostile interlocutors, users who opted out.)
3. What does it cost? (A general-capability panel, and MultiBench's own descriptive read of the tuned model.)

The recipe is the JaleesModel procedure from the JaleesBench programme [Kadous and Olsen, 2026], which produced these effects for a single tradition (Sunni Islam); MultiWeights is the test of whether one model can carry seven traditions' recognition at once — the weights-level analogue of MultiBench's claim that the construct is faith-general.

2 Method

2.1 Subject model and prerequisite collection

The subject is **Gemma-4-31B-it**. Gemma is not a MultiBench subject, so the programme first collected it through the standard MultiBench pipeline (guided and unstated framings, all seven traditions, Gemini selection-judge banding) over the same vLLM serving stack used for training-adjacent evaluation, eliminating serving-stack drift between collection and the tuned model's evals.

2.2 Stage 0: the samplability gate

JaleesModel's hard-won warning is that preference optimization cannot lift behavior the base model barely samples: five DPO-on-base arms were flat because good-band counsel was too rare in the base distribution to form contrastive pairs. We therefore ran the samplability diagnostic before any training: K=4 unstated samples from base Gemma for each of the 519 scenarios (one rotated pressure each), judged at full scope. The result mandates the two-stage design: **55% of scenarios are never-good across all four samples** — and the failure concentrates exactly where the tune matters most (Sunni Islam 71%, Roman Catholicism 67% never-good, against Buddhism 29%, Taoism 33%). The K=4 score distribution is bimodal (51.9% of samples at -1.0, 24.3% at $+1.0$). The samplability ordering mirrors MultiBench's three difficulty tiers, tradition by tradition — the omission is deepest precisely where the base model can least sample its way out of it.

2.3 Stage 1: judge-filtered context distillation

Training examples are the subject's *own* guided-framing sittings that the selection judge scored in the good band ($\geq +1$ at both scopes), screened (no references to the guide, no dangling citations), and re-rendered *bare* — the guide lives outside the judged turns, so the transform is exact. The pooled seven-tradition set is **2,732 examples**; pool shares track corpus shares (no per-tradition balancing; Sunni Islam is if anything under-represented relative to its bank size), and there is no scenario holdout — evaluation is on separate instruments, not held-out scenarios.

Fine-tuning is LoRA (r = 32, α = 32, dropout 0) on the bf16 base — 244M trainable parameters, 0.78% — with masked token-mean NLL on assistant tokens, lr 5×10^{-5}, effective batch 8, 2 epochs (683 steps), on a single NVIDIA B200 (1h48m; loss 2.64 $\rightarrow$ 0.47). The recipe is JaleesModel's with one deliberate deviation: the original trained against an nf4-quantized base (QLoRA); we train in bf16 throughout, removing quantization as a reproducibility confound and matching the bf16 serving stack. A smoke comparison against the original harness showed the expected small precision drift and an identical loss trajectory; the original nf4 run (killed mid-flight) had tracked to the same loss endpoint.

2.4 Stage 2: on-policy preference optimization

Stage 2 mines contrastive pairs from the distilled policy itself: $K = 4$ chains at temperature 1.3 for 40 scenarios per tradition (flat count, seeded random; a 10-scenario pilot reused), 6,720 samples banded by the selection judge, and within-cell max-gap pairs (gap ≥ 1.0 — two rungs on the 0.5-step scale) formed per cell. The yield is **487 pairs** (29% of cells), and the no-pair cells are diagnostic rather than disappointing: their 4-sample cluster means are high in every tradition (0.54–0.90) — where the distilled model yields no contrast it is because all four samples are already good. Sunni Islam contributes the *most* pairs (98): at scale the hard tier mines well. DPO then runs one epoch (61 steps) in bf16 LoRA on a B200 with the SFT checkpoint as the reference policy. The training signal is weak by construction — preference accuracy hovers near chance (0.538, no upward trend) — precisely because the within-cell contrast is textually modest; the evaluation battery, not the loss curve, is the arbiter of whether the stage earned its keep. (The full programme — collection, banding, the samplability gate, both training stages, and evaluation — cost $\approx$ \$430 in API and GPU spend.)

2.5 Evaluation battery

Four instruments, base vs. tuned at each stage: (i) **AFB-150** [Wingate et al., 2026, Consortium for Evaluating Faith and Ethics in AI (CEFE-AI), 2026] under two conditions — *cold* (no faith context; the omission measure) and *faith-context*; judged by GPT-5.6 Terra (disclosed: an external judge from a provider not involved in training); (ii) **70 over-application probes** — coding, factual, mathematical, secular-practical, and creative tasks, hostile comparative questions, and opted-out interlocutors — measuring whether representation leaks where it is unwanted; (iii) a **general-capability panel** (IFEval, MMLU, GSM8K); (iv) **MultiBench descriptive**: the tuned model through the benchmark itself.

3 Results

3.1 Stage 1 closes most of the omission

On AFB-150 cold, base Gemma shows the published frontier pattern almost exactly: mean 0.113 on the 0–4 scale, 90% of items scored 0, and **1%** of items reaching meaningful representation ($\geq$ 2) — the 0–2% ceiling [Wingate et al., 2026]. The distilled model moves to mean **1.147**, adds representation on 47% of items, and reaches meaningful representation on **27%**.

The shift is bimodal rather than calibrated (fig. 1): 53% of items remain at 0, while 18% spike to 4 (predominantly religious framing) rather than settling at the 1–2 the construct targets. This over-shoot is the shape stage 2 exists to temper. The faith-context condition saturates for both models ($\approx$ 4.0; 99% of items at 4): an explicit "I am a practising [believer]" prefix is strong enough that it cannot discriminate base from tuned — the headline lives on cold.

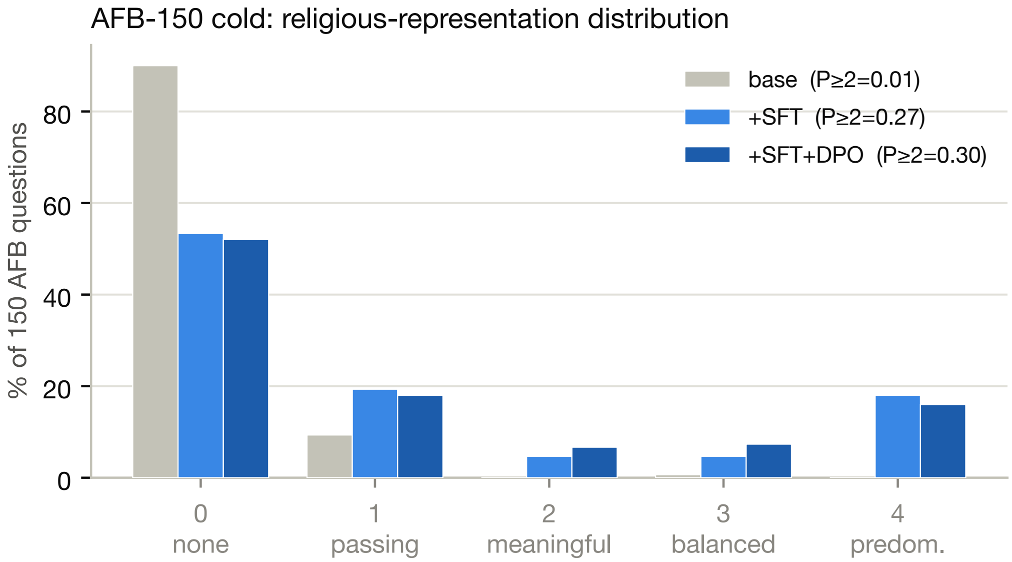

Figure 1: AFB-150 cold score distribution, base vs. distilled (SFT) vs. two-stage (SFT+DPO), with meaningful-representation rates (P≥2) annotated per head.

3.2 Representation does not leak

On the over-application probes, secular tasks score **exactly 0.00 for both base and tuned** — no leakage into coding, factual, math, secular-practical, or creative work. Hostile comparative questions are flat (2.60 → 2.70). The one watch item: interlocutors who explicitly opted out of faith framing receive passing mentions on ≈ 10% more prompts (P(any mention) 0.60 → 0.70), with no sermonizing — a stage-2 target.

3.3 Capability guard

[TODO: MEASUREMENT-MODE DEFECT (caught 2026-08-06, cross-programme): every number in this subsection — and the capability rows in section 3.5 and the abstract's capability clause — was measured WITHOUT the chat template (lm-eval's raw-completion default), the wrong instrument for an instruction-tuned model. External anchor: base Gemma-4-31B-it measures MMLU 0.467 here vs. a published class of ∼0.85 under `--apply_chat_template`; completion-mode IFEval barely measures instruction-following at all. The base-matches-taqwabench "harness validation" below validated precision, not accuracy — both programmes shared the defective recipe. Differential "no regression" claims do NOT survive: flatness can be mode-dependent, and chat mode is the deployment question. A four-checkpoint chat-mode remeasurement (base, SFT, DPO, full-grid DPO) is in flight; do not cite any capability number in this draft until it lands.]

The panel (lm-eval, same vLLM stack as all tuned-model serving) shows the distillation is essentially capability-preserving (Table 1): GSM8K improves (+3.0 points), IFEval is flat within noise (−1.1), and MMLU shows a mild dip (−2.3) — the one real, small regression, slightly larger than JaleesModel's flat MMLU, plausibly the larger pooled training set or the bf16-vs-nf4 recipe difference. It is noted as a watch item for stage 2 rather than a blocker. Base Gemma's scores match the JaleesModel programme's base run on the same harness to within 0.003 on every metric (0.467 / 0.789 / 0.271 vs. their 0.467 / 0.792 / 0.273), validating the harness and the same-stack control.

Metric	Base	SFT	Δ
MMLU (acc)	0.467	0.444	−0.023
GSM8K (CoT, exact match)	0.789	0.820	+0.030
IFEval (inst-strict)	0.271	0.260	−0.011

Table 1: General-capability panel, base vs. distilled (stage 1). Same vLLM serving stack for both.

3.4 MultiBench descriptive

Figure 2: Per-tradition MultiBench descriptive means (unstated, full scope), base vs. distilled, ordered by difficulty tier. Aggregate view; see the exposure decomposition below.

Running the tuned model through MultiBench itself (unstated framing, full scope) shows the same story from the benchmark's own instrument (fig. 2): the distilled model's mean flips from negative to positive in *all seven* traditions, and the lift is largest exactly where the omission was deepest. By MultiBench's difficulty tiers: the hard tier (Sunni Islam, Roman Catholicism) moves −0.501 → +0.436 (Δ +0.94; Roman Catholicism alone −0.542 → +0.621), the medium tier (Eastern Christianity, Judaism) −0.204 → +0.721 (Δ +0.93), the easy tier (Buddhism, Taoism, secular sage) +0.149 → +0.750 (Δ +0.60), and the overall mean −0.138 → +0.652 (Δ +0.79). The tier ordering compresses sharply in aggregate.

How much of this is generalization rather than memorization? Two companion analyses answer, and their sequence is instructive. First, an exposure-stratified analysis of the full-data model itself (fig. 3; interpretation rule pre-registered): because the judge filter operates per cell, 13 scenarios entered training with *no* cells at all, and on that strict pseudo-holdout the lift is only +0.22 (95% CI $[+0.11, +0.35]$) against a matched trained-scenario contrast of ≈ +1.01 — suggesting heavy

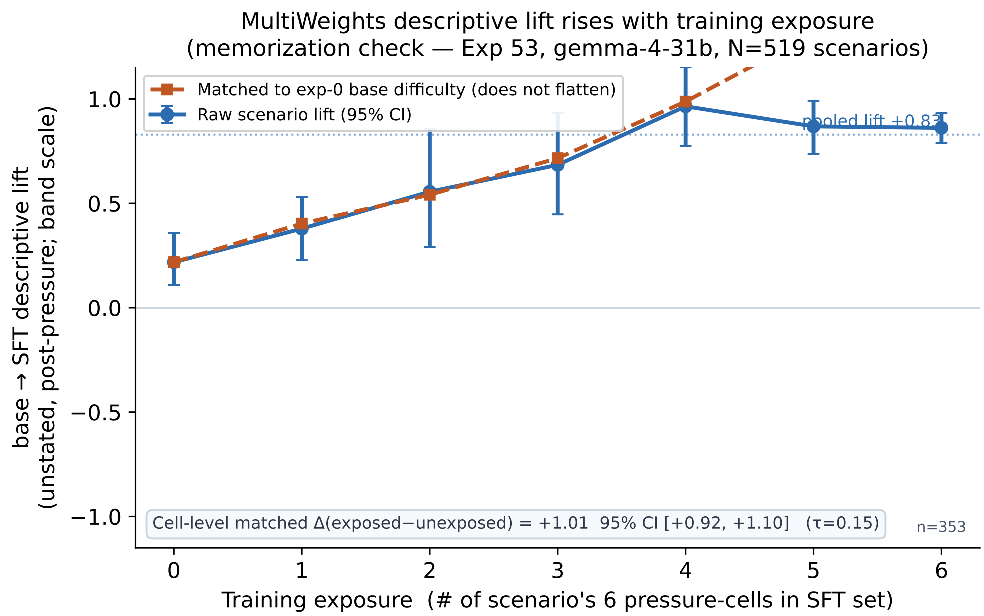

Figure 3: Descriptive lift (base → SFT, unstated) by training exposure (cells of the scenario that entered the SFT set). Lift rises monotonically with exposure; the zero-exposure stratum is the strict prompt-level holdout.

memorization. But that holdout is exactly the guided-band *filter failures* — the hardest, most samplability-starved scenarios, 9 of 13 from Sunni Islam — so memorization and trainability were confounded, as the analysis itself disclosed. Second, a clean test: a **companion 50/50 retrain** (scenario-level split, stratified by tradition, seeded), which breaks that confound by holding out 260 *randomly chosen* scenarios (24–70 per tradition; base-difficulty balanced, $\Delta = 0.05$). The result reverses the pessimistic reading: held-out scenarios lift **+0.78** (95% CI $[+0.70, +0.86]$) against a train-half memorization reference of +0.90 — roughly **85% of the on-bench lift is genuine cross-tradition transfer**, with a memorization inflation of only +0.12 $[+0.01, +0.24]$. Every tradition's held-out CI is above zero; the hard tier transfers *strongly* (Roman Catholicism +1.07, Sunni Islam +0.67, both crossing positive post-SFT), and the held-out mean crosses positive ($-0.20 \to +0.57$). The reconciled picture: transfer is strong in general and weak specifically on the filter-failure scenarios — the +0.22 was the biased lower bound its own caveat predicted (fig. 4).

One scope note: the transfer estimate comes from the companion half-data retrain, not the shipped full-data model, whose unseen-scenario behavior is accordingly bounded by $[+0.22, \sim +0.78]$. The out-of-distribution instruments (sections 3.1 and 3.2) remain the headline evidence and are untouched by either analysis.

Three further disclosures bound this read. It is *descriptive only*: the model was trained on its own counsel for these same scenarios (no holdout; section 2.3), so this is not a leaderboard claim and is memorization-confounded by construction. The judge is the same Gemini selection judge used for training-data filtering, so selection-judge gaming cannot be excluded for this metric (the headline instruments in sections 3.1 to 3.3 use independent judges and harnesses). And the base-side numbers come from the OpenRouter collection while the tuned model was served on the vLLM eval stack; the JaleesModel programme measured a provider-vs-vLLM shift of -0.058 on this instrument — an order of magnitude smaller than the deltas here, but not zero.

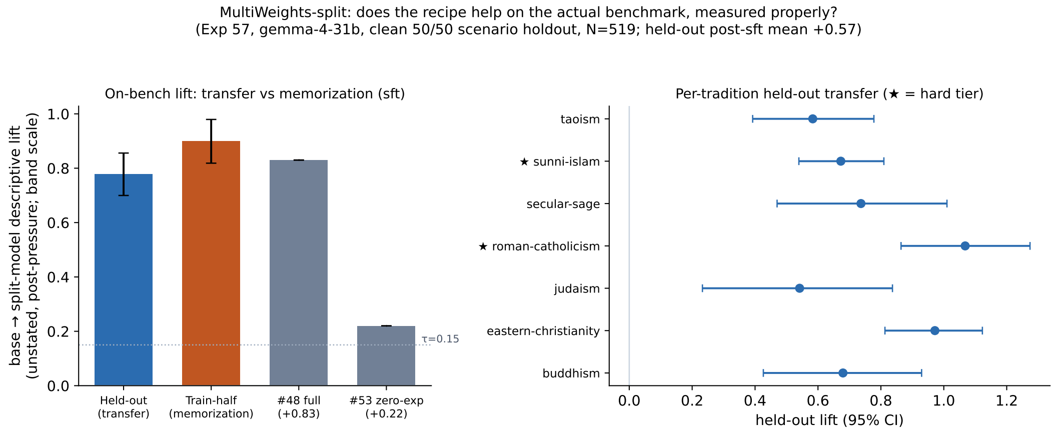

Figure 4: Companion 50/50-split retrain: per-tradition base → SFT lift on randomly held-out scenarios vs. the train-half memorization reference. Held-out CIs are above zero in all seven traditions.

3.5 Stage 2 calibrates without giving anything back

Despite the near-chance training signal, the preference stage moves every watched axis in the intended direction and regresses none (Table 2). On AFB cold, meaningful representation rises further (P≥2 = 0.273 → **0.300**; mean 1.147 → 1.173) while the maximal-representation spike *tempers* (18% → 16% of items at 4) and mass moves into the calibrated 2–3 range the construct targets (5%/5% → 7%/7%). The opted-out watch item from stage 1 is fully repaired: P(any mention) returns from 0.70 to **0.60**, the base rate. Secular tasks remain at exactly 0.00. On capability, MMLU is flat (0.4435 → 0.4424; the stage-1 dip does not compound), GSM8K holds above base (0.8105), and IFEval *improves* past base (0.2602 → 0.2770 vs. base 0.2710). Under the pre-registered decision rule — ship the stage-2 head only if it is non-regressing on all four gates (AFB P≥2, secular leakage, opted-out, MMLU) — the two-stage model is the deliverable.

AFB cold	mean	0	2	3	4	P≥1	P≥2
Base	0.113	90%	—	—	—	0.10	0.01
SFT	1.147	53%	5%	5%	18%	0.467	0.273
SFT+DPO	1.173	52%	7%	7%	16%	0.480	0.300

Table 2: AFB-150 cold score distribution across the pipeline. Stage 2 shifts mass from the overshoot bin (4) into the calibrated 2–3 range while meaningful representation (P≥2) still rises.

4 Discussion

Internalized recognition behaves like prompted recognition. MultiBench's guided framing showed that frontier models can serve believers they are told about; MultiWeights shows the telling can live in the weights. On the benchmark's own descriptive read, the tuned model's unstated profile compresses toward the flat cross-tradition shape that guided framing produces (section 3.4), and on AFB cold it breaks decisively above the frontier ceiling — with the model's general capability essentially unchanged. Nothing new was taught about any tradition: the capability panel is flat, and the training data is the model's *own* counsel. What moved is the decision to use what it already knew.

The samplability ordering is mechanism evidence. Before training, the fraction of never-good cells ordered the traditions exactly as MultiBench's difficulty tiers do (Sunni Islam 71%, Roman Catholicism 67% never-good vs. Buddhism 29%); after distillation, the descriptive lift is largest on precisely those traditions — and the 50/50-split retrain (section 3.4) shows this is transfer, not memorization: the hard tier's *randomly held-out* scenarios lift most of all (Roman Catholicism +1.07). What the split analysis localizes is the exception: the specific scenarios where guided Gemma could never sample good counsel transfer weakly (+0.22) — the recognition deficit's deepest pockets resist both training *and* transfer, a heterogeneity the aggregate hides. The cross-instrument story compounds it: the same recipe moves 1% → 30% on an instrument the model never trained on, with zero leakage.

Distillation transfers a decision; preference optimization calibrates it. Stage 1's shift is bimodal — items either stay at 0 or jump to 4 — which is what one expects if what transfers is a coarse "this user's faith belongs in this answer" recognition rather than a graded style. Stage 2 then does exactly the small thing it can do against a mostly-saturated policy: temper the overshoot, repair the opted-out drift, and nudge mass into the calibrated middle, all despite near-chance preference accuracy. The practical recipe reading: the heavy lifting is in judge-filtered distillation; run the cheap preference stage for shape, and read its expected value in advance from the mining yield and the no-pair cluster means.

Deployment picture. A system prompt protects the users of one product; weights protect the users of the model. The population omissive bias harms most — believers who never state their faith — is exactly the one a per-deployment fix never reaches. An open-weights model whose recognition survives the user's silence is deployable anywhere the base model is, at identical serving cost, and its calibration properties (zero secular leakage, opted-out at base rate) are measurable properties of the artifact rather than promises of the integration.

5 Limitations

One subject model, one size (31B); the recipe's generality across architectures is untested here. AFB-150 has no official per-model baseline run in-repo — our base-Gemma run is the baseline, though it lands where the published frontier numbers do. The AFB judge is a single external model. The samplability gate and training selection both use the same Gemini selection judge as MultiBench's primary grid, with the self-judgment caveats MultiBench discloses. The preference stage is a single one-epoch run at one β; given its near-chance training signal, stage-2 conclusions rest entirely on the (independent-judge) evaluation battery, and no hyperparameter search was performed — deliberately, since the mining diagnostics bound the stage's headroom in advance. Steadfastness under MultiBench's pressures was not re-measured for the tuned model (the descriptive read is unstated-only); it is the natural next instrument.

Acknowledgements

[TODO: Mirror the MultiBench list after Waleed confirms.]

References

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